



The impact of visual crowding on Chinese character recognition: a comparative study of primary school children and adult university students

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Abstract

Visual crowding is considered a primary factor limiting reading speed and accuracy. Nevertheless, research on how visual crowding affects the recognition of Chinese characters, especially from a developmental perspective, remains scarce. To address this gap, the present study compared the impact of visual crowding on Chinese character recognition among primary school children and adult university students. Experiment 1 showed significantly higher character recognition accuracy in the uncrowded condition compared to the crowded condition, indicating that crowding significantly affected Chinese character recognition. Furthermore, adults demonstrated significantly higher recognition accuracy than children in both small and large target-flanker separations, suggesting that children are more susceptible to visual crowding than adults. Experiments 2 and 3 further explored the impact of visual crowding by varying character attributes. The results revealed that these attributes significantly affected character crowding, and that adults' recognition accuracy was higher than children's across all font types and sizes. Notably, no significant interaction was observed between the participant group and these attributes, indicating that children and adults were similarly affected by these attributes. The present study reveals the differential effects of crowding on Chinese character recognition among children and adults, which can provide valuable insights for optimizing character attributes in reading materials to mitigate the impact of visual crowding and improve reading efficiency.

Keywords Visual crowding · Chinese character recognition · Peripheral vision · School-aged children · Character attributes

Introduction

Reading is one of the most critical avenues for acquiring knowledge, and the accuracy of Chinese character recognition directly influences reading efficiency. During the reading process, most texts fall within the peripheral vision, which may trigger the visual crowding effect, significantly affecting character recognition. Visual crowding refers to the phenomenon where the recognition of a target object is greatly disrupted, or even fails, when surrounded by flanking objects. This effect is prevalent in various visual

activities, including reading (Gong & Smart, 2021; Levi, 2008; Whitney & Levi, 2011). Research has shown that the processing speed of texts under visual crowding conditions is significantly slower compared to when presented in isolation (Dobres et al., 2018). Furthermore, the loss of peripheral letter information due to visual crowding accounts for 75% of all losses, far exceeding factors such as peripheral low visual acuity (Yu et al., 2014). This indicates that visual crowding has a substantial impact on text recognition (Beier & Oderkerk, 2021b; Cheung & Cheung, 2017) and is therefore considered a major factor limiting reading speed and accuracy (Dobres et al., 2018; Hakvoort et al., 2017; Joo et al., 2018).

The impact of visual crowding on text recognition can be influenced by many factors. First, the intensity of crowding increases with the eccentricity of the target (Song et al., 2014; Strappini et al., 2017), which refers to the distance

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between the target stimulus and the fixation point, typically quantified in degrees of visual angle. Second, according to Bouma's law (Bouma, 1970), the degree of crowding depends on target-flanker separation (e.g., Whitney & Levi, 2011). At a given eccentricity, crowding decreases as the distance between the target and the flanking objects increases (Astle et al., 2014; Gurnsey et al., 2011; Marten-Ellis & Bedell, 2021). Research on reading has also shown that reading speed increases with the spacing between letters (Chiu & Drieghe, 2023; Yu et al., 2007). Moreover, visual crowding in text is also influenced by font attributes such as font type (Bachmann & Mengheri, 2018; Beier & Oderkerk, 2021a; Bernard et al., 2016; Galliussi et al., 2020), font size (Huurneman et al., 2016; Lages et al., 2017; O'Brien et al., 2005; Zhu et al., 2021), font weight (Bernard et al., 2013; Dobres et al., 2016b), and font complexity (Bernard & Chung, 2011; Thiessen et al., 2022). For instance, Chinese characters with more strokes and more complex structures tend to produce more pronounced visual crowding than simpler characters (Zang et al., 2016; Zhang et al., 2009). Moreover, an increased in font size (Alotaibi, 2007; O'Brien et al., 2005) or weight (Beier & Oderkerk, 2021a; Dobres et al., 2016a) typically results in a reduction of visual crowding effects.

However, previous research on visual crowding in text has primarily focused on alphabetic scripts, which are relatively simple visual stimuli with relatively consistent spatial complexity across words (Grainger et al., 2008; Wiley et al., 2016; Zhang et al., 2009). In contrast, Chinese characters possess complex visual structures, making the impact of visual factors more significant during reading (Cole & Pickering, 2010; McBride-Chang et al., 2011). Meanwhile, due to the presence of characters with varying spatial complexities flanking individual characters, Chinese characters tend to suffer from more visual crowding compared to alphabetic scripts (Pelli et al., 2006; Wang et al., 2014). Furthermore, crowding in characters occurs not only between characters but also within them (Li et al., 2019; Zhang et al., 2009). Despite this, research investigating the impact of visual crowding on Chinese characters lags significantly behind that of alphabetic scripts like English. When the structures of characters are simple, they are less affected by crowding (Guo et al., 2018; Wang et al., 2014). Additionally, under visual crowding conditions, the accuracy of target character recognition with smaller inter-character spacing is significantly higher than that with larger spacing, indicating that inter-character spacing significantly affects the performance of children with dyslexia in Chinese reading (Meng et al., 2020).

Furthermore, previous research on the visual crowding of Chinese characters has primarily focused on adults (Korinth et al., 2020; Siman-Tov et al., 2021) or children with dyslexia

(Bertoni et al., 2019; Meng et al., 2020), with a scarcity of studies on normal children. While the examination of visual crowding in children with abnormalities is of considerable clinical significance, research on typically developing school-age children is equally crucial. This population comprises the vast majority of school-age children, who are in a critical stage of developing their Chinese reading ability, characterized by transitioning from being "beginner readers" to "skilled readers." Compared to adults, their visual perception are more susceptible to low-level factors (Shen et al., 2010). Additionally, primary school children experience crowding in the fovea (Jeon et al., 2010), whereas adults typically exhibit no or minimal crowding in the fovea (Kalpadakis-Smith et al., 2022). Norgett and Siderov (2014) found that children aged 4 to 6 and 7 to 9 suffered from higher degrees of crowding than adults with a mean age of 25 years. This is largely attributed to the immaturity of their visual system (Doron et al., 2020; Garey, 1984).

In summary, while studies have shown that visual crowding has a significant impact on Chinese character recognition (Lyu et al., 2022; Wang et al., 2020), research on crowding of Chinese characters is insufficient. Notably, there remains a dearth of research on the differences in the impact of visual crowding on Chinese character recognition between "less skilled" child readers and "skilled" adult readers. To address these gaps, we designed three experiments. In Experiment 1, we explored the differential effects of crowding on Chinese character recognition between children (primary school students) and adults (university students). It was hypothesized that compared to adults, children's reading ability would be more susceptible to the effects of visual crowding. In Experiments 2 and 3, we further explored how character attributes influence visual crowding in Chinese characters, and whether these attributes differentially modulate the crowding effects on character recognition across the two participant groups. Specifically, in Experiment 2, we manipulated font type and size, while in Experiment 3, we manipulated font complexity and weight to investigate the impact of Chinese character attributes on crowding across both group. We hypothesized that SimHei would be less susceptible to visual crowding compared to SimSun, KaiTi, and FangSong. We also hypothesized that larger font sizes would be less affected by visual crowding than smaller sizes, and characters with lower complexity and thicker strokes would be easier to recognize. These hypotheses were confirmed by the current study. In other words, we showed that children were more susceptible to visual crowding than adults in Chinese character recognition. Altering character attributes like font type, size, complexity, and weight affected visual crowding and recognition accuracy in both primary and university students. These findings provide a theoretical guidelines for optimizing typographic design in

reading materials, suggesting that visual crowding during reading can be effectively reduced, and reading efficiency can be improved by adjusting Chinese character properties such as font type, size, and weight. Additionally, personalized reading guidance should also be provided based on readers' developmental stage. Since children are more susceptible to visual crowding than adults when recognizing Chinese characters, greater emphasis should be placed on optimizing the properties of Chinese characters in children's reading materials. This can reduce the potential impact of visual crowding on them and improve their character recognition efficiency.

Experiment 1

Method

Participants

A total of 55 participants were recruited for this experiment. Among them, 30 were primary school students, including 16 lower-grade students (9 males, 7 females, with a mean age of 7.75 ± 0.77 years) and 14 upper-grade students (6 males, 8 females, with a mean age of 10.79 ± 1.19 years). Adult participants consisted of 25 university students, including 14 males and 11 females, with a mean age of 20.12 ± 1.92 years. All participants were native Chinese speakers with normal or corrected-to-normal vision, color vision, and right-handedness. None of them had participated in similar experiments before, and they were unaware of the experimental purpose prior to the experiment.

Stimuli

The Chinese characters used in the experiments were selected from the vocabulary list of the lower-grade Chinese textbooks compiled by the Ministry of Education of China. This version of textbooks are the most prevalent adopted in primary schools throughout China. Crucially, they are also used by the participants in our study. Subsequently, five primary school Chinese teachers assessed the selected characters to ensure that they had been learned by and were familiar to the participants. Ultimately, a database of 200 Chinese characters was finalized for experimental purpose.

The complexity of the characters is determined by the number of strokes. Specifically, low complexity refers to characters with 3 to 5 strokes, medium complexity refers to characters with 6 to 9 strokes, and high complexity refers to characters with 10 to 12 strokes. In Experiment 1, the target characters were of medium complexity, while the distractor characters had either 5 or 10 strokes. The font size of the

characters was set to 24 points (0.93° in visual angle). In the no flanker condition, only a single target character was presented. In the flanker condition, three characters were presented simultaneously, with the middle character being the target and the flanking characters serving as distractors. The distance between the characters varied according to the level of crowding. In the small target-flanker (TF) separation condition, the target character was positioned adjacent to the distractor characters, with a character-to-character visual angle of 1.08° . In the large target-flanker separation condition, the distractor and target characters were separated by one space, resulting in a character-to-character visual angle of 1.75° .

Design and procedure

A three-factor mixed design was employed, consisting of 2 (participant group: primary school students, university students) $\times$ 3 (target-flanker separation: small, large, no flanker) $\times$ 2 (eccentricity: 3° , 6°). Participant group was a between-subject variable, while target-flanker separation and eccentricity were within-subject variables. When the eccentricity was 3° , the target characters were presented in the parafoveal region of the eye, while at 6° , they were presented in the peripheral region. The dependent variable was the accuracy of target character recognition.

Our experimental procedure was adapted from classic visual crowding paradigm, which has also been employed by other researchers to investigate the phenomenon of crowding between Chinese characters (e.g., Meng et al., 2020). Stimulus presentation and response acquisition were implemented using E-prime 2.0 (Psychology Software Tools Inc.). During the experiment, participants were seated approximately 60 cm away from the computer screen, with the Chinese character stimuli presented in black font on a white background. In each trial, instructions were first displayed on the screen, which were then explained to the participants by the experimenter. Afterward, the experiment began by pressing "p" on the keyboard. A fixation cross "+" was displayed at the center of the screen for 1000 ms. Following this, a triplet of characters comprising a target character flanked by two other characters on either side (crowding condition) or a single target character (uncrowded condition) was presented at the fixation point location for 150 ms. Subsequently, a recognition display with four characters appeared. Among these four, one was the target character, two were the previously presented flanker characters, and the last one was a randomly selected character. Participants were required to click on the target stimulus using the left mouse button.

The experiment consisted of 252 trials, including 12 practice trials and 240 experimental trials. The formal experiment was presented in three blocks based on the

target-flanker separation. The experimental progress was self-paced, but participants were forced to take a break after every 40 trials. The entire experiment took approximately 25 min to complete.

Results and discussion

Data were analyzed using IBM SPSS 26. A three-way repeated measures Analysis of Variance (ANOVA) was performed on the accuracy rate of Chinese character recognition. The results revealed that the three-way interaction among participant group, target-flanker separation, and eccentricity was not significant, $F(2, 106)=1.44$, $p=0.24$.

A significant interaction was observed between participant group and target-flanker separation, $F(2, 106)=17.53$, $p<0.001$, $\eta_p^2=0.25$. Specifically, under the condition of small target-flanker separation, adults' accuracy rate in Chinese character recognition ($M=0.79$, $SD=0.1$) was significantly higher than that of children ($M=0.64$, $SD=0.2$), $p<0.001$. Similarly, under the condition of large target-flanker separation, adults' accuracy rate ($M=0.83$, $SD=0.1$) was also significantly higher than that of children ($M=0.66$, $SD=0.2$), $p<0.001$. Yet, under the no flanker condition, there was no significant difference in the accuracy rate between adults ($M=0.94$, $SD=0.1$) and children ($M=0.94$, $SD=0.1$), $p=0.73$ (see Fig. 1a). Moreover, a significant interaction was also found between target-flanker separation and eccentricity, $F(2, 106)=11.18$, $p<0.001$, $\eta_p^2=0.17$. When characters were located at the eccentricity of 3° , the accuracy rate for the no flanker condition ($M=0.96$, $SD=0.1$) was significantly higher than that under the large target-flanker separation condition ($M=0.79$, $SD=0.2$) and the small target-flanker separation condition ($M=0.77$,

$SD=0.2$), $ps<0.001$, while no significant difference was found between the large and small target-flanker separation conditions, $p=0.167$. For characters presented at the eccentricity of 6° , the accuracy rate for the no flanker condition ($M=0.92$, $SD=0.1$) was significantly higher than that under the large ($M=0.69$, $SD=0.2$) and small ($M=0.65$, $SD=0.2$) target-flanker separation condition, $ps<0.001$. Furthermore, the accuracy rate under the large target-flanker separation condition was significantly higher than that under the small target-flanker separation condition, $p=0.02$ (see Fig. 1b). The interaction between participant group and eccentricity was not significant, $F(1, 53)=1.69$, $p=0.20$.

The main effects of participant group, target-flanker separation and eccentricity were significant, $F(1, 53)=17.78$, $p<0.001$, $\eta_p^2=0.25$, $F(2, 106)=136.00$, $p<0.001$, $\eta_p^2=0.72$ and $F(1, 53)=61.73$, $p<0.001$, $\eta_p^2=0.54$, respectively.

In summary, Experiment 1 revealed that visual crowding had a significant detrimental effect on Chinese character recognition among both children and adults. Notably, adults demonstrated significantly higher recognition accuracy than children under the crowded conditions, whereas no such difference was observed in the uncrowded condition. This result indicated that children were more significantly affected by visual crowding when recognizing characters compared to adults. Previous studies have shown that character attributes can impact character crowding (e.g., Meng et al., 2020). However, since character attributes were not manipulated in this experiment, the extent to which they differentially modulate crowding effects on character recognition between children and adults remains unclear. To address this gap, Experiment 2 was designed to investigate this question.

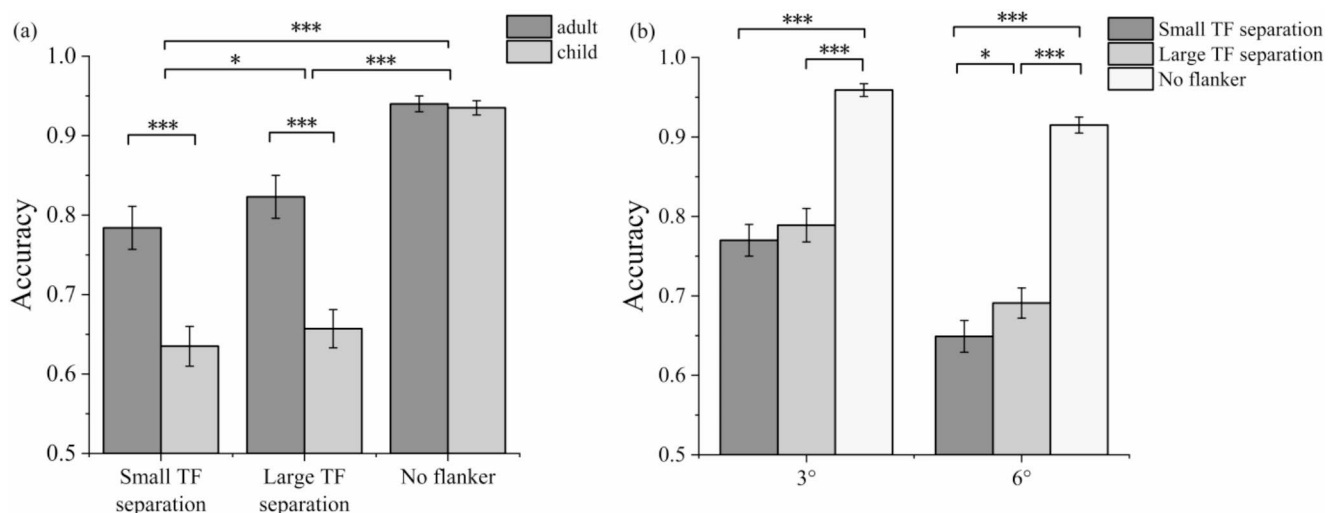


Fig. 1 Results of experiment 1. *Note.* The two bar graphs depict the interaction between (a) target-flanker separation and participant group, and (b) target-flanker separation and eccentricity. TF separation repre-

sents target-flanker separation. Error bars represent one standard errors of the mean. * $p<0.05$, *** $p<0.001$

Experiment 2

Method

Participants

A total of 58 participants were recruited for this experiment. Among them, 28 were primary school students, including 17 lower-grade students (8 males, 9 females, with a mean age of 7.82 ± 0.64 years) and 11 upper-grade students (3 males, 8 females, with a mean age of 10.64 ± 0.81 years). Additionally, 30 were university students, consisting of 18 males and 12 females, with a mean of 20.23 ± 1.78 years. All participants were native Chinese speakers with normal or corrected-to-normal vision, color vision, and right-handedness. None of them had participated in similar experiments before, and they were unaware of the experimental purpose prior to the experiment.

Stimuli

The Chinese characters used in the experiment were drawn from the database created in Experiment 1. Both the target and flanker characters were the same as those utilized in Experiment 1. The target character was positioned at an eccentricity of 5° . The flanker characters matched the target character in attributes such as font type, size, and boldness. Additionally, the visual angle between the target and flanker was set at 1.08° .

Design and procedure

A three-way mixed design was adopted, consisting of 2 (participant group: primary school students, university students) $\times$ 4 (font types: FangSong, SimHei, KaiTi, SimSun) $\times$ 3 (font sizes: 18, 24, 34). The four fonts were selected because they are most commonly used in primary school textbooks. Specifically, textbooks for first and second grades primarily use KaiTi font, while textbooks for third and fourth grades gradually transition from KaiTi to SimSun, and by grades 5 to 9, SimSun becomes the dominant font. In addition to KaiTi and SimSun, the main fonts used also include SimHei (often used for titles, emphasized texts, and illustrations), and FangSong (often used for subtitles, poems, annotations, citations, etc.). According to the “Preventing and Controlling Myopia Work Plan for Primary and Middle School Students” issued by the Ministry of Education of China in 2008, the recommended eye-to-book distance for primary and middle school students is one inch (32–35 cm). In this experiment, the distance between the participants and the computer was approximately twice this distance, so the three different font sizes corresponded to 9-point, 12-point,

and 17-point fonts on the book, which are commonly used font sizes in different reading materials. The participant group was the between-subject variable, while font type and font size were the within-subject variables. The dependent variable was accuracy of character recognition.

The experimental procedure was identical to Experiment 1.

Results and discussion

A three-way repeated measures ANOVA was conducted. The results revealed that the three-way interaction between participant group, font size, and font type was not significant, $F(6, 336)=0.95$, $p=0.46$. The two-way interaction between participant group and font type was not significant, $F(3, 168)=0.64$, $p=0.59$. The interaction between participant group and font size was also not significant, $F(2, 112)=0.19$, $p=0.83$.

However, a significant interaction was found between font size and font type, $F(6, 336)=2.847$, $p=0.01$, $\eta_p^2=0.048$. Specifically, when the font size was 18 points, the recognition accuracy for the fonts of SimHei ($M=0.76$, $SD=0.2$), SimSun ($M=0.73$, $SD=0.2$), and KaiTi ($M=0.71$, $SD=0.2$) was significantly higher than that of FangSong ($M=0.65$, $SD=0.2$), $ps<0.01$. Additionally, the recognition accuracy for SimHei was also significantly higher than that of KaiTi, $p=0.036$, but no significant differences were found between SimHei and SimSun, or between KaiTi and SimSun, $p=0.17$ and $p=0.38$, respectively. When the font size was 24 points, there were no significant differences in recognition accuracy among the various font types. For font size 34 points, the recognition accuracy for SimHei ($M=0.87$, $SD=0.1$) was significantly higher than that of FangSong ($M=0.82$, $SD=0.2$), KaiTi ($M=0.82$, $SD=0.2$), and SimSun ($M=0.83$, $SD=0.0$), $ps<0.05$, while no significant differences were observed between other font types (see Fig. 2).

The main effect of participant group was significant, $F(1, 56)=44.99$, $p<0.001$, $\eta_p^2=0.45$, indicating that the accuracy of Chinese character recognition was significantly higher for adults ($M=0.86$, $SD=0.1$) than for children ($M=0.69$, $SD=0.2$). Also, both the main effect of font size and that of font type were significant, $F(2, 112)=100.55$, $p<0.001$, $\eta_p^2=0.64$ and $F(3, 168)=13.59$, $p<0.001$, $\eta_p^2=0.20$, respectively.

In summary, Experiment 2 revealed a significant interaction between font type and font size, along with significant main effects of these two attributes, suggesting that font type and font size significantly influence crowding of Chinese characters. However, no significant interactions were observed between participant type (children versus adults) and these font attributes, indicating that the influence of font type and size on character crowding is comparable across

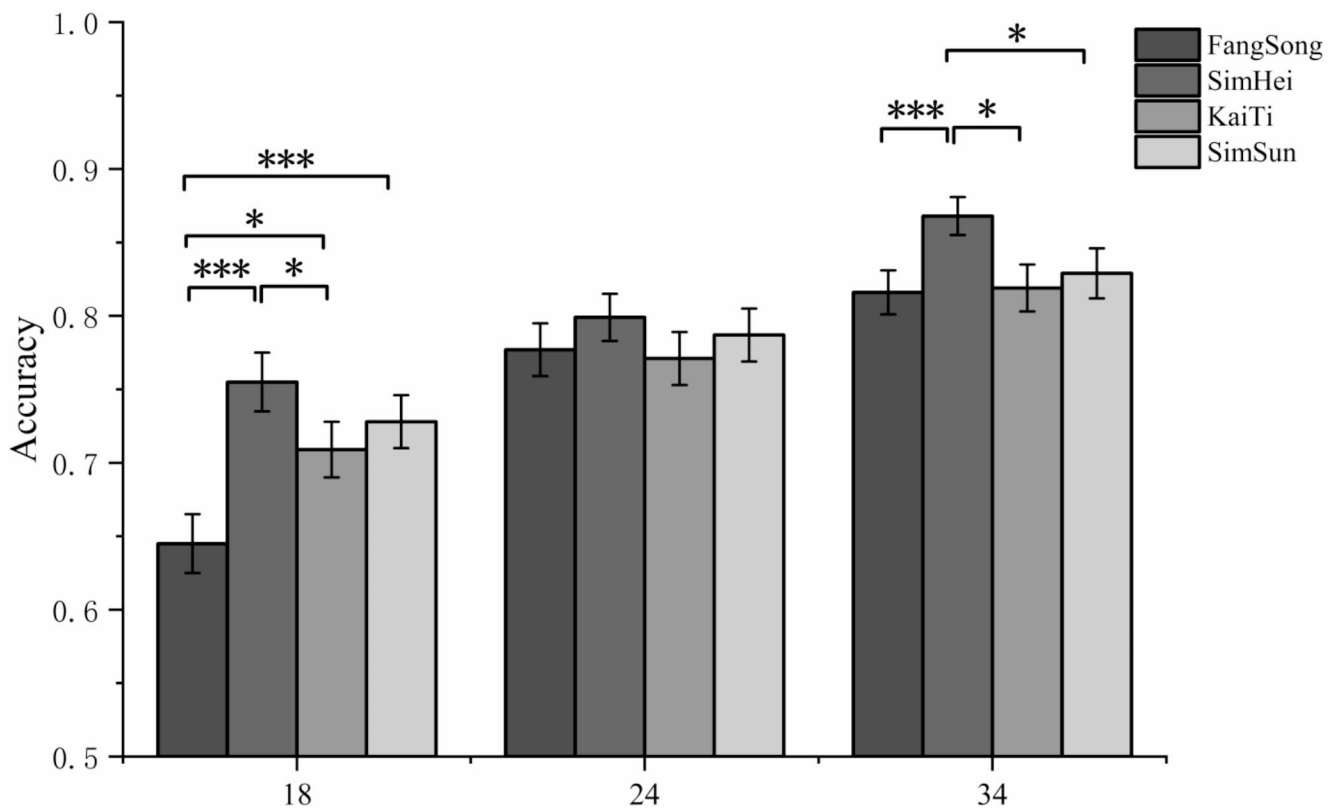


Fig. 2 Mean accuracy under different conditions in experiment 2. *Note.* Error bars represent one standard errors of the mean. * $p < 0.05$, *** $p < 0.001$

age groups. In fact, apart from font type and font size, other attributes such as text complexity (Zang et al., 2016; Zhang et al., 2009) and font weight (Beier & Oderkerk, 2021a; Dobres et al., 2016a) also affect text crowding. In the next experiment, we further investigated whether other font attributes, including boldness and complexity, affected character crowding and whether they modulated character crowding differently across children and adults.

Experiment 3

Method

Participants

A total of 58 participants were recruited for this experiment. Among them, 30 were primary school students, including 15 lower-grade students (8 males, 7 females, with a mean age of 9 ± 0 years) and 15 upper-grade students (8 males, 7 females, with a mean age of 12 ± 0 years). Adult participants consisted of 28 university students, including 14 males and 14 females, with a mean age of 20.71 ± 2.00 years. All participants were native Chinese speakers with normal or corrected-to-normal vision, color vision, and right-handedness. None of them had participated in similar experiments

before, and they were unaware of the experimental purpose prior to the experiment.

Stimuli

The Chinese characters used in this experiment were also from the database established in Experiment 1. The font size of both the target and the flanker characters was 24 points, and the target character was positioned at an eccentricity of 5° . The flankers were of medium complexity, and their font type, size, bolding, and other attributes were consistent with the target characters. The distance between the target and the flankers was 1.08° in visual angle.

Design and procedure

A 2 (participant group: primary school students, university students) $\times$ 2 (font type: KaiTi, SimSun) $\times$ 2 (font complexity: low, high) $\times$ 2 (font weight: bold, normal) mixed design was adopted. The participant group was a between-subject variable, while the other three were within-subject variables. The dependent variable was accuracy.

The experimental procedure was the same as in Experiment 1. It consisted of 300 trials, including 12 practice trials and 288 formal experimental trials. The entire experiment took approximately 30 min to complete.

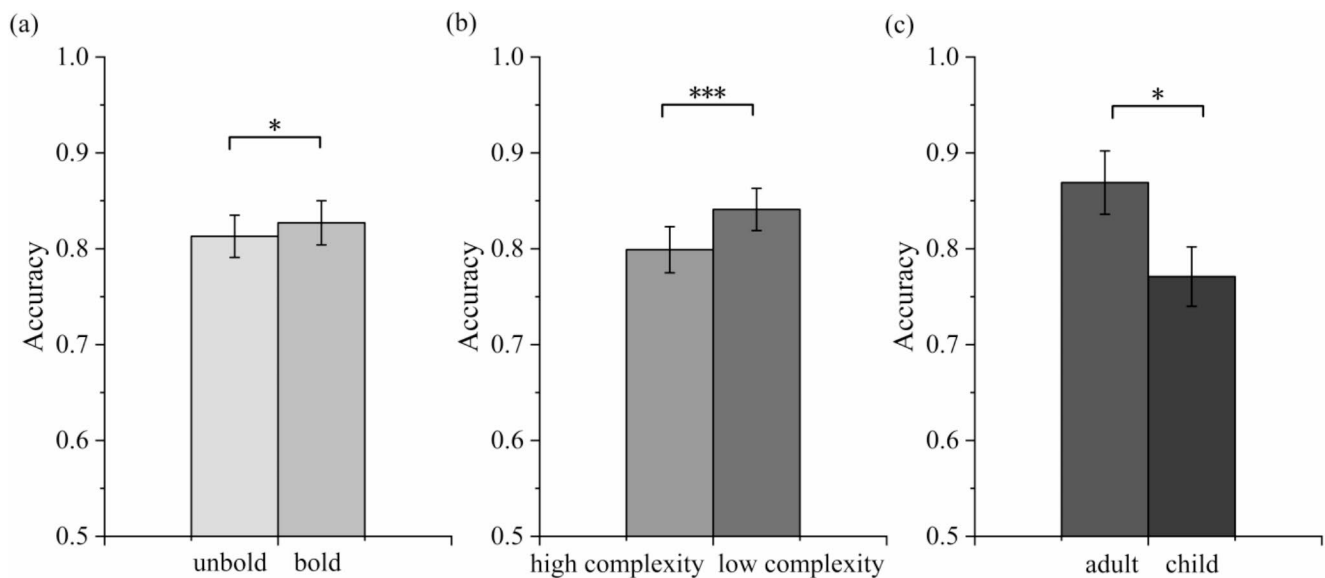


Fig. 3 Mean accuracy of Chinese character recognition under each level of (a) font weight, (b) font complexity, and (c) participant group. *Note.* Error bars represent one standard errors of the mean. * $p < 0.05$, *** $p < 0.001$

Results

A four-factor repeated measures ANOVA was performed on the accuracy of character recognition revealed a significant main effect of font weight, $F(1,56) = 5.6$, $p = 0.022$, $\eta_p^2 = 0.1$. The accuracy for bold fonts ($M = 0.82$, $SD = 0.2$) was significantly higher than for normal fonts ($M = 0.81$, $SD = 0.2$) (see Fig. 3a). A significant main effect of font complexity was also observed, $F(1,56) = 37.2$, $p < 0.001$, $\eta_p^2 = 0.4$, with low-complexity fonts ($M = 0.84$, $SD = 0.2$) being better recognized than high-complexity fonts ($M = 0.8$, $SD = 0.2$) (see Fig. 3b). The main effect of the participant group was also significant, $F(1,56) = 4.654$, $p = 0.035$, $\eta_p^2 = 0.08$. The recognition accuracy of Chinese characters among college students ($M = 0.87$, $SD = 0.1$) was significantly higher than that among primary school students ($M = 0.77$, $SD = 0.24$) (see Fig. 3c). Other main effects and interactions were not significant.

General discussion

The present study examined the impact of visual crowding on Chinese character recognition among normal primary school students and university students. The results indicated that compared with adults, children's recognition of characters was more susceptible to the effects of visual crowding, regardless of whether the effect was strong or weak. Furthermore, for both participant groups, altering the properties of Chinese characters, such as font type, font size, font complexity, and font weight, could affect visual crowding, thereby influencing the accuracy of character recognition.

Intriguingly, primary school students and university students were similarly modulated by these attributes. These three experiments demonstrated that under the same visual presentation conditions, children with lower reading proficiency were more significantly impacted by visual crowding during Chinese character recognition. This verifies the previous conclusion that visual crowding is developmental (Jeon et al., 2010).

Experiment 1 revealed that, under the no-flanker condition, there was no significant difference in the accuracy of Chinese character recognition between adults and children. This suggests that, in the absence of peripheral visual stimulus interference, children's visual perception abilities are roughly equivalent to adults. However, under both large and small target-flanker separation conditions, adults demonstrated significantly higher accuracy in recognizing Chinese characters compared to children, highlighting children's more vulnerable character recognition abilities in visually crowded environments. This result aligns with previous findings suggesting an age-related enhancement in the processing of ambiguous visual information (Balas et al., 2020; Doron et al., 2020). Norgett and Siderov (2014) discovered that children with an average age of 5 years old performed more poorly in visual crowding task compared to adults. The authors suggested that this discrepancy could be attributed to the incomplete retinal or cortical development of children. The retina and cortical structure from the age of 4 to 6 are continuous developing and maturing (Lewis & Maurer, 2005). Furthermore, during object identification, the visual system must actively suppress distractors (Crook et al., 1998; Priebe & Ferster, 2008). However, children's lateral inhibition ability has not fully matured (Amestoy et

al., 2021; Gwiazda et al., 1997). This developmental lag can hinder their capacity to effectively eliminate interference from surrounding objects during object discrimination, resulting in visual crowding. Consequently, the immature visual system of children makes them more susceptible to interference from surrounding information when processing complex visual inputs (Atkinson et al., 1986; Doron et al., 2020; Norgett & Siderov, 2014). Additionally, the development of children's reading skills can also influence their performance in character crowding task. As children mature, their Chinese character recognition skills (Li et al., 2020) and reading proficiency (Hui et al., 2018; Xue et al., 2013) improve, as well as their cortex and inhibition ability (e.g., Amestoy et al., 2021). Consequently, they experience reduced interference from visual crowding.

Experiment 1 also revealed that under eccentricities of 3° and 6°, the recognition accuracy under the no-flanker condition was significantly higher than that under the flanker condition, regardless of the separation between target and flankers. This indicates that crowding reduces the accuracy of Chinese character recognition in both the parafoveal (3°) and peripheral (6°) regions. At the eccentricity of 6°, a significant difference in accuracy was observed between the large and small target-flanker separation conditions, whereas no such difference was noted at 3° eccentricity. This can be attributed to the retina's higher density of cone cells at small eccentricities, which are crucial for spatial information processing. This enables the visual system to still effectively recognize the target at even small target-flanker separation condition. Thus, there is no significant difference between the impacts of small and large target-flanker separations on visual crowding. As eccentricity increases, peripheral visual acuity diminishes (Strasburger et al., 2011), leading to a more pronounced crowding effect, as demonstrated by the present study and many others (e.g., Livne & Sagi, 2007). At larger eccentricities, the change of target-flanker separation can more significantly affect crowding. At greater eccentricity, an enlarged target-flanker separation facilitates the segregation of the target character from the flanker, leading to better performance outcomes (Buonocore et al., 2017). Thus, the change in target-flanker separation in the condition of small eccentricity that does not lead to a significant change in crowding may cause a significant change in crowding in the condition of large eccentricity. The lack of significant interaction between participant group and eccentricity suggests that eccentricity affects Chinese character recognition similarly for both children and adults, regardless of their visual development or learning experience.

In Experiment 2, the impact of font size on the accuracy of Chinese character recognition was observed. As the font size increased, the recognition became more precise. This finding aligns with previous research (Alotaibi, 2007; Bai

et al., 2011; Zhu et al., 2008). Chinese character recognition adheres to the principles of graphical identification—under constant spatial frequency conditions, larger fonts facilitate easier recognition of Chinese characters (Zhu et al., 2008). As font size increases, the complex and slightly crowded strokes and components of the characters become more distinct, making both holistic and partial processing easier, which leads to enhanced accuracy in recognition. However, some studies argue that font size does not affect recognition efficiency (Russell & Chaparro, 2001). This discrepancy might arise because the subjects of this experiment were Chinese characters, not alphabetic scripts. Chinese characters possess higher visual complexity and are more susceptible to visual factors (McBride-Chang et al., 2011), resulting in more significant visual crowding (Lyu et al., 2022). Chinese characters under crowded conditions tend to be perceived as more blurred stimuli compared to alphabetic scripts. Therefore, the enlargement of font size leads to a more significant improvement in perceptual accuracy.

Experiment 2 also discovered that, regardless of font size, adults and children's recognition accuracy was not lower for SimHei compared to SimSun, KaiTi and FangSong. The recognition advantage of SimHei characters can be attributed to their unique typographic characteristics. Characters in SimHei are thicker, simpler and have more regular strokes (Dobres et al., 2016). As a sans-serif font, SimHei has minimal decoration at the beginning and end of strokes, and the thickness of strokes is uniform (Casas-Tost & Rovira-Esteva, 2018). This facilitates separating multiple components of Chinese characters and identifying the font structure. Consequently, the recognition speed is faster than serif fonts such as KaiTi and FangSong (Zhang et al., 2014). Furthermore, in Experiments 2 and 3, there was no significant difference in recognition accuracy between SimSun and KaiTi. As two common Chinese fonts, SimSun originates from KaiTi, so there is a certain similarity between them (Wu, 2017). However, Gong et al. (2009) found that participants could recognize SimSun faster than KaiTi. Hence, even though both SimSun and KaiTi can be accurately recognized in reading, there is a difference in recognition speed that could influence reading experience. Therefore, the impact of font on recognition speed should be taken into consideration when typesetting reading materials.

Experiment 3 of the present study revealed that bold Chinese characters exhibited a higher recognition accuracy, suggesting that they were less affected by visual crowding compared to non-bold ones. Similar findings have been reported in studies on alphabetic scripts (Bernard et al., 2013). However, Dobres et al. (2016a) reported differing findings. They conducted a study involving female participants with an average age of 37 years and male participants with an average age of 44.2 years, finding no significant

difference in recognition accuracy between bold and regular fonts. This discrepancy may be due to the fact that they used characters with a height of 5 mm, while the characters used in this experiment were 8.66 mm tall. Larger characters may be easier to distinguish in terms of their details, thereby increasing the recognition difference between bold and regular fonts. This discrepancy may be attributed to the typically more acute visual systems of school-aged children and young adults (Calabrèse et al., 2016). As individuals age, the acuity of their visual system may decrease, leading to a weakened ability to distinguish between bold and regular fonts.

Visual crowding increases with the complexity of letters (Bernard & Chung, 2011). Experiment 3 demonstrated that this principle also applies to Chinese characters, as characters with low complexity were recognized more accurately than those with high complexity. For Chinese characters, the complexity in experimental settings is typically gauged by the number of strokes, with a greater number of strokes indicating higher visual complexity (Zang et al., 2016). The crowding in Chinese characters occurs not only between characters but also within characters (Zhang et al., 2009). For high-complexity characters, the strokes are more likely to intertwine both within- and between-characters, leading to stronger within-character crowding and between-character crowding. As a result, complex Chinese characters require more attention for visual processing and exhibit a lower probability of being skipped during reading (Liversedge et al., 2014).

Limitations and future directions

There are several limitations to this study. Firstly, the target Chinese characters in this study were flanked by only one distractor character on each side, whereas in real-life reading scenarios, characters often appear in paragraphs, surrounded by other characters in all directions. This could potentially lead to stronger crowding effects and worse recognition performance. Therefore, the ecological validity of this study is limited. Future research may assess character crowding by presenting the target character within scenarios that closely mimic real-life contexts. Secondly, although the results of the present study did not reveal a statistically significant interaction between font attributes and participant group, this does not imply that font type and size have no impact on Chinese character recognition. Future studies could incorporate additional indicators, such as reaction time or even electrophysiological measures, to comprehensively evaluate their influence. Thirdly, the current study only examined the character crowding in primary school children and adult university students, thus providing an incomplete developmental perspective on visual crowding.

Subsequent research could more systematically delve into the developmental trajectory of the influence of visual crowding on Chinese character recognition. For instance, it would be insightful to identify at which stage (e.g., middle school or high school) individuals experience similar levels of visual crowding as adults. Furthermore, future investigation could test the effectiveness of the optimal Chinese character attributes for character recognition. Should the findings affirm that the optimal attributes, as identified in the present study, do enhance reading efficiency, textbooks and supplementary reading materials could incorporate these optimized font attributes for typesetting in the future.

Conclusion

The present study demonstrates that both children and adults' recognition of Chinese characters is affected by visual crowding, but children are more affected. Additionally, character recognition during Chinese reading is modulated by attributes such as font size, font type, font weight, and character complexity. Specifically, the larger the font size, the less impact visual crowding has on Chinese characters; the font of SimHei is less affected by visual crowding than KaiTi, SimSun, and FangSong. Additionally, bold characters and low complexity characters are less influenced by visual crowding. However, children and adults were similarly affected by these attributes.

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Data availability The data and analyses for the experiments reported here are available from the Open Science Framework: https://osf.io/s7829/?view_only=71098ed70b7e4a5d99e7c3fb0e693ca2.

Declarations

Consent to participate Informed consent was obtained from all individual participants included in the study.

Competing interests The authors report there are no competing interests to declare.

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